

★ Make Sure you are caught up on videos and readings. Material becomes twice as hard

LECTURE 12: MATH1061/7861  
TOPICS: MODULAR ARITHMETIC  
DATE: THURSDAY MARCH 19, 2026

next week.

Learning objectives:

- (1) Understand the Quotient-Remainder Theorem.
- (2) Understand how to perform modular arithmetic.
- (3) Understand the definition of the floor and ceiling of a number.

★ Class Rep (MATH 1061)

$$\rightarrow \forall n, d \in \mathbb{Z}, d \neq 0$$

unique  $\rightarrow \exists ! q, r \in \mathbb{Z}$  such that

$$n = \underbrace{q}_{\text{quotient}} d + \underbrace{r}_{\text{Remainder}} \quad \text{and} \quad \underline{0 \leq r < |d|}$$

$$\textcircled{1} \quad 6 = 3 \times 2 + \underline{0}$$

Remainder.

$$\textcircled{2} \quad -7 = \underbrace{(-4)}_{\text{quotient}} \times 2 + \underbrace{(1)}_{\text{remainder}}$$

$$\textcircled{3} \quad -2 = (1) \times (-7) + 5$$

$a$  is congruent to  $b$  modulo  $d$ .

$\leadsto a \equiv b \pmod{d}$  means  $d \mid (a-b)$ .

Q1: True or false:

T (1)  $\exists x \in \mathbb{N}$  such that  $3x \equiv 1 \pmod{5}$

Take  $x=2$ . Then note  $5 \mid (3 \times 2 - 1)$

$$x=7 \Rightarrow 3 \times 7 - 1 = 20 \quad \checkmark$$

$$x=12 \Rightarrow 3 \times 12 - 1 = 35 \quad \checkmark$$

Any  $x$  of the form  $5K+2$  works,  $K \in \mathbb{Z}$ .

F (2)  $\exists x \in \mathbb{N}$  such that  $3x \equiv 1 \pmod{6}$

There is no such  $x$ .

Indeed, if  $x$  is even, then  $x = 2K$ .

Then  $3x = 6K$  is divisible by 6.

So cannot have remainder 1 modulo 6.

If  $x$  is odd, then  $x = 2K+1$ .

$3x = 6K+3$ . This has remainder 3

modulo 6, so cannot have remainder 1.

Another argument: Suppose, for the sake of contradiction,  $\exists x \in \mathbb{N}$  s.t.  $3x \equiv 1 \pmod{6}$ .

$$\Rightarrow 6 \mid (3x-1) \Rightarrow 3 \mid (3x-1)$$

$$\Rightarrow 3 \mid (-1) \quad \text{contradiction} \quad \square$$



$\lfloor x \rfloor = \text{biggest integer} \leq x.$

Q2: True or false:

$$\lfloor 2.5 \rfloor = 2$$

$$\lfloor -2.5 \rfloor = -3$$

False.

$$\forall x, y \in \mathbb{Q}, \lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$$

$$x = 0.6 \quad y = 0.7$$

$$\lfloor x + y \rfloor = \lfloor 1.3 \rfloor = 1$$

$$\lfloor x \rfloor + \lfloor y \rfloor = 0 + 0 = 0.$$

True or False:  $\lfloor x + y \rfloor$  is either equal to  $\lfloor x \rfloor + \lfloor y \rfloor$  or to  $\lfloor x \rfloor + \lfloor y \rfloor + 1$ .

Example:

$$x = -0.5 \Rightarrow \lfloor x \rfloor = \lfloor -0.5 \rfloor = -1$$

$$y = -0.4 \Rightarrow \lfloor y \rfloor = \lfloor -0.4 \rfloor = -1$$

$$\lfloor x + y \rfloor = \lfloor -0.9 \rfloor = -1$$

$$\lfloor x + y \rfloor = -1 = \underbrace{\lfloor x \rfloor + \lfloor y \rfloor}_{-2} + 1$$

Q2 Proof that  $\lfloor x+y \rfloor$  is either equal to  $\lfloor x \rfloor + \lfloor y \rfloor$  or  $\lfloor x \rfloor + \lfloor y \rfloor + 1$ :

As  $\lfloor x \rfloor$  is the smallest integer less than or equal to  $x$ ,

$$x = \lfloor x \rfloor + a \quad \text{for some } a \in \mathbb{R} \text{ with } 0 \leq a < 1.$$

Likewise,  $y = \lfloor y \rfloor + b$  for some  $b \in \mathbb{R}$  with  $0 \leq b < 1$ .

As  $\lfloor x \rfloor + \lfloor y \rfloor$  is an integer,

$$\begin{aligned} \lfloor x+y \rfloor &= \lfloor (\lfloor x \rfloor + a) + (\lfloor y \rfloor + b) \rfloor \\ &= \lfloor (\lfloor x \rfloor + \lfloor y \rfloor) + (a+b) \rfloor \\ &= \lfloor x \rfloor + \lfloor y \rfloor + \lfloor a+b \rfloor. \end{aligned}$$

Because  $0 \leq a < 1$  and  $0 \leq b < 1$ , we know that  $0 \leq a+b < 2$

and hence  $\lfloor a+b \rfloor$  is either 0 or 1.

This means that either  $\lfloor x+y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor + 0$  or

$$\lfloor x+y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor + 1.$$

□

Q3:

(a) Prove that for all odd integers  $n$ , we have  $n^2 \equiv 1 \pmod{4}$ .

Let  $n$  be an odd integer, so  $n = 2k+1$  for some  $k \in \mathbb{Z}$ .

$$\text{Then } n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 4(k^2 + k) + 1.$$

Thus  $n^2$  has remainder 1 when divided by 4,

$$\text{and so } n^2 \equiv 1 \pmod{4}.$$

□

(b) Use part (a) to show

$$\left\lfloor \frac{n^2}{4} \right\rfloor = \left\lfloor \frac{n^2 - 1}{4} \right\rfloor. \quad (\text{for all odd integers } n)$$

~~We will consider 2 cases~~

~~We will break up the problem~~

Take any odd integer  $n$ . From part a, we know that

$$4 \mid (n^2 - 1), \quad \text{so } n^2 - 1 = 4k \text{ for some } k \in \mathbb{Z}.$$

$$\text{Then LHS} = \left\lfloor \frac{n^2}{4} \right\rfloor = \left\lfloor \frac{n^2 - 1 + 1}{4} \right\rfloor = \left\lfloor k + \frac{1}{4} \right\rfloor = k,$$

as  $k$  is an integer.

$$\text{Also RHS} = \left\lfloor \frac{n^2 - 1}{4} \right\rfloor = \left\lfloor \frac{4k}{4} \right\rfloor = k, \quad \text{therefore } \left\lfloor \frac{n^2}{4} \right\rfloor = \left\lfloor \frac{n^2 - 1}{4} \right\rfloor$$